



Cryptography

We often use cryptography while communicating between two devices via the internet. This helps us secure the information from malicious threats.



We live in an age where we all share and receive data. Data is in the form of texts, emails, photos, videos, live streams, and so on. We constantly share some kind of information through various means.

A problem with information flow is that it can be intercepted and read by someone other than the intended recipient. This is important because we often share private information and confidential messages, intended for specific people. To prevent interception, we need a secure transmission method. Data stealing can expose us to many different types of cyber-attacks.

- **Phishing**

A fraudulent practice of sending mass emails to people to trick them into sharing private or sensitive information. Usually phishing is via mail and commonly asks for personal details including banking information. This may lead to identity theft, financial scams, etc.

- **Man in the Middle**

In an MITM attack, the hacker tracks or redirect packets sent from a user to a recipient.

Cryptography

We are all familiar with the concept of sign languages and traffic signals used on streets. They are basically gestures or images with information hidden in them. Storing information inside something else is called encoding. Understanding that information is decoding.

Cryptography is the science of encoding and decoding information in order to safeguard its integrity.

Why is cryptography necessary?

- a. Confidentiality: To protect confidential information and safely communicate between two users.
- b. Data Privacy: To safeguard private and sensitive data from potential attackers.
- c. Authentication: To function as an authorization layer for the right people to access data. It also functions as an identifier.
- d. Security – To function as a layer of security and safeguard data from threats.

Cryptosystems

Even before the age of computers, hidden messages have been seen throughout history and popular culture. Spies, armies, and pirates used this method to secretly deliver messages.

Exercise

 Decode the following codes - Each has a different logic

Cipher:

- 1** CVVCEM CV FCYP (Key: Shifting letters by 2 spaces to the right for example C converts to A)

 _____

- 2** TAERGEHTREDNAXELA (Key: Reverse the letters)

 _____

The Caesar Code

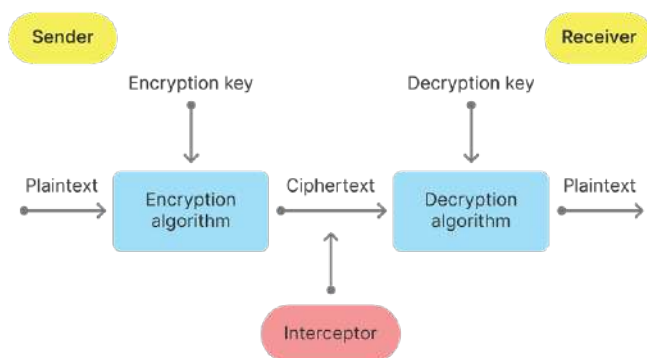
The Caesar Code is one of the oldest methods to create a cipher. Originally used by Julius Caesar, the logic behind the code was to shift the letters 3 steps to the right to make the cipher. The letters will be adjusted 3 steps left to retrieve the original message.

From the simple shifting of codes, to the use of symbols and machines, we have come a long way. Cryptography is an ancient art which has led to a historic turn of events.

Cryptosystem

Cryptosystem is a system in which these messages are encoded and decoded.

The classical cryptosystem model that has been followed is:



Let us understand the roles of a cryptosystem's various components.

- **Plaintext**

The message we intend to send to the recipient.

- **Encryption**

Encryption is the process to apply algorithms to a message and convert it into unintelligible text or hide it inside another message.

- **Encryption Algorithm**

A mathematical process that produces a ciphertext from a given plaintext, using an Encryption Key.

- **Cipher Text**

After encrypting the data, we get a "ciphertext". Ciphertext is a completely unintelligible piece of information. If someone reads it without knowing the encryption method, they will not know what the message is.

- **Interceptor**

The third person who tries to read the data we are sending.

- **Decryption**

The process of converting encrypted data (cipher text) into its original message.

- **Decryption Algorithm**

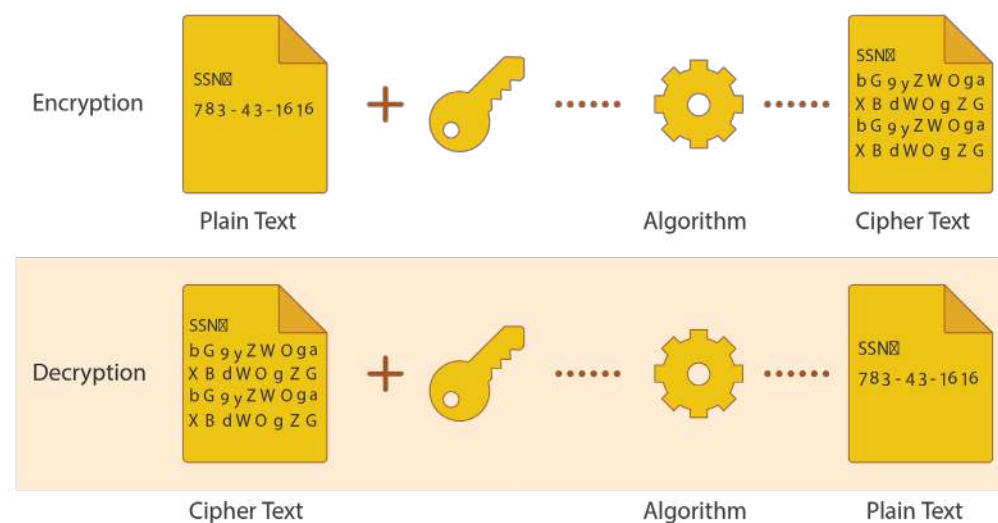
A mathematical process which produces plaintext from the given ciphertext, using a decryption key.

- **Encryption Key**

A value or table of agreed rules which is entered into the encryption algorithm to make a cipher text
The sender knows the encryption key.

- **Decryption Key**

The value known to the receiver.



Activity 1

Create your own cipher

There are no strict rules for algorithms to make a cipher text, as long as both parties are aware of how it was coded in the first place.

Message: **“Class at seven”**

The algorithm used to encrypt this text is – ROT13, where we shift each alphabet by 13 places.

LETTERS	A	B	C	D	E	F	G	H	I	J	K	L	M
CODE	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
LETTERS	N	O	P	Q	R	S	T	U	V	W	X	Y	Z
CODE	A	B	C	D	E	F	G	H	I	J	K	L	M

C	L	A	S	S	A	T	S	E	V	E	N
P	Y	N	F	F	N	G	F	R	I	R	A

Code – “PYNFFNGFRIRA”

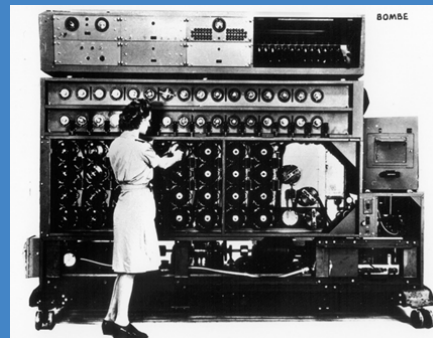
Other methods to create ciphers are:

- Reverse the text and add numbers between the letters.
- Change text to symbols
- Change text to numbers with mathematical operations.

Make a cipher with your friend and share messages with each other. Decrypt them to find out what the message was.

Bytes

In the mid-20th century, during its early years, scientists developed machines that allowed communication with others through encrypted messages. These machines were known as “Enigma,” and their encryption code was referred to as “the Enigma code.” Sir Alan Turing, considered the father of modern computing, played a crucial role in deciphering the Enigma code. He and his team devised the Turing Machine, which enabled scientists to break the code by implementing basic algorithms for decryption. By utilizing a Turing Machine, any given algorithm could be simulated, signifying a significant milestone in the development of the modern computers we use today. This was the one of the major pioneering work towards the invention of modern computers.



Alan Turing is known as the Father of Computer Science and Artificial Intelligence.

Types of cryptosystems

There are two types of cryptosystems based on the manner in which encryption and decryption is carried out in the system:

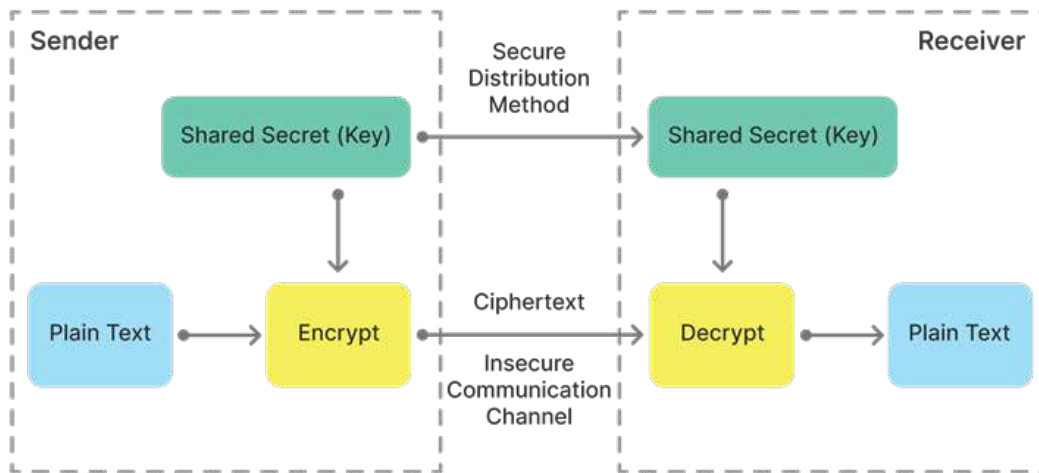
1. Symmetric Key Encryption
2. Asymmetric Key Encryption

The main difference between these cryptosystems is the relationship between the encryption and the decryption key. Logically, in any cryptosystem, both keys are closely associated. It is practically impossible to decrypt the ciphertext with a key that is unrelated to the encryption key.

Symmetric Key Encryption:

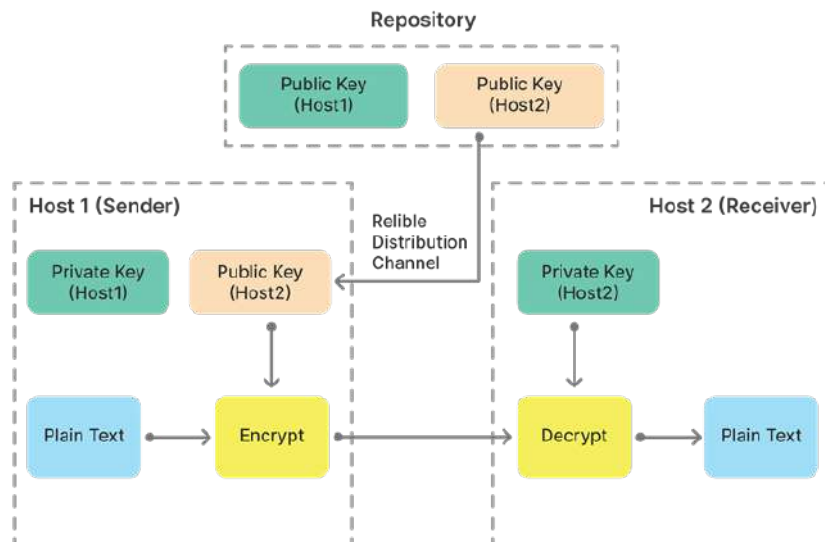
An encryption process where the same keys are used to encrypt and decrypt information is known as Symmetric Key Encryption.

The study of symmetric cryptosystems is referred to as symmetric cryptography. Symmetric cryptosystems are also called secret key cryptosystems.



Asymmetric Key Encryption:

The encryption process where different keys are used to encrypt and decrypt information is known as Asymmetric Key Encryption. Though the keys are different, they are mathematically related. So, retrieving the plaintext by decrypting the ciphertext is feasible. The process is depicted in the following illustration:



Here is how Asymmetric Key Encryption works:

- Every user in the system needs to have a pair of dissimilar keys: a private key and a public key. These keys are mathematically related. When one key is used for encryption, the other can decrypt the ciphertext back to the original plaintext.
- The public key is put in a public repository and the private key is kept a well-guarded secret. This encryption scheme is also called Public Key Encryption.
- When User1 needs to send data to User2, he obtains User2's public key from the repository, encrypts the data, and transmits it. User2 uses his private key to extract the plaintext.
- The key length (number of bits) in this encryption is large. So, the encryption-decryption process is slower than symmetric key encryption.

Symmetric cryptosystems are a natural concept. In contrast, public-key cryptosystems are quite difficult to understand.

Attribute	Symmetric	Asymmetric
Keys	One key is shared between two or more than two entities	One entity has a public key and another has the corresponding private key
Key Exchange	Out-of-band through secure mechanisms	Public key is available to everyone while private key is kept secret by the owner
Speed	The algorithm is less complex and faster	Algorithm is more complex and slower
Use	Bulk encryptions like communication paths	Digital signatures and key distribution
Security Provided	Confidentiality	Authentication and nonrepudiation

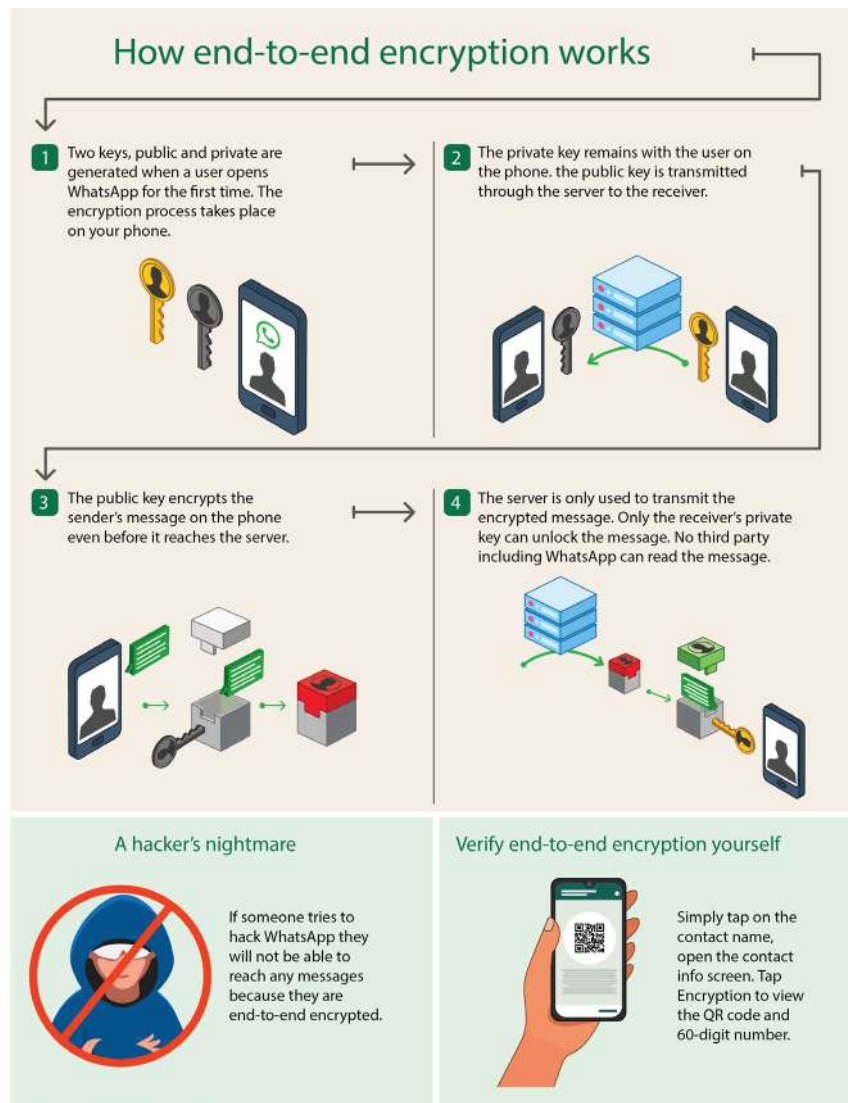
Bytes

Out of band: A 2-factor authentication system. For example, integrating a phone number with an account number which requires a One Time Password (OTP) to log in.

Nonrepudiation: It means that to the ability to ensure that a device in communication cannot deny the authenticity of their signature on a document or the sending of a message that they originated.

Applications of cryptography

In our daily lives, cryptography is most commonly used in messaging applications. Most messaging apps are end-to-end encrypted. This means that only the sender and receiver can view its contents, because only they have the public and private key needed to open the message packets.



Process of sending message from user 1 to user 2

Online banking transactions and emails are end-to-end encrypted and follow the same methodology, sometimes with additional layers of security. They could require one-time-passwords (OTP), biometric scans, etc.

UPI app payment services also use encryption, since it is a transaction.

Exercise

Fill in the blanks

- 3 _____ is one of the oldest methods of creating a cipher.
- 4 A mathematical process that uses an encryption key to produce a ciphertext from a given plaintext is known as an _____ .
- 5 UPI app payment services also use _____ .

State whether True or False

- 6 The system in which messages are encoded and decoded is the Cryptosystem.
▲ _____
- 7 The encryption process where the same keys are used for encrypting and decrypting information is Asymmetric Key Encryption.
▲ _____
- 8 Cryptography is widely and most commonly used in our daily lives in messaging applications.
▲ _____



Tech Flash

- **Cryptography** : Cryptography is the science of encoding and decoding information in order to safeguard its integrity.
- **Encryption** : Encryption is the process of applying algorithms to our message which converts it into unintelligible text or hides it inside another message.
- **Encryption Algorithm** : A mathematical process that produces a ciphertext, from a given plaintext, using an encryption key.
- **Ciphertext** : A ciphertext is what we get after encrypting the data.
- **Interceptor** : The third person who is trying to read the data we send.
- **Decryption** : The process of converting encrypted data (cipher text) into its original message (plaintext).
- **Decryption Algorithm** : A mathematical process which produces plaintext from the given ciphertext using a decryption key.
- **Encryption Key** : It is a value or table or agreed rules which are entered into the encryption algorithm so as to make cipher text. It is known to the sender.
- **Decryption Key** : It is the value known to the receiver.
- **Symmetric Key Encryption** : An encryption process where the same keys are used to encrypt and decrypt information.
- **Asymmetric Key Encryption** : An encryption process where different keys are used to encrypt and decrypt information.



Future Tech

Cryptocurrency

Cryptocurrency is a digital asset which does not rely on banking systems. It works as a medium of exchange, substituting currencies which we traditionally use.



What is Cryptocurrency?



Cryptocurrency is digital money created from code



The Cryptocurrency economy is monitored by a peer-to-peer internet protocol.



Cryptocurrency is an encrypted string of data or a hash, encoded to signify one unit of currency